Formula & Trends Sheet

CBSE Class 12 Chemistry • NCERT Part I, Chapters 1–5

Blue boxes: NCERT syllabus. Purple boxes: JEE extras.

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Constants

R	8.314 J K^{-1} mol^{-1}	F	96487 $\approx$ 96500 C mol^{-1}
R	0.083 L bar K^{-1} mol^{-1}		
K_f (water)	1.86 K kg mol^{-1}	K_b (water)	0.52 K kg mol^{-1}
K_f (benzene)	5.12 K kg mol^{-1}	K_b (benzene)	2.53 K kg mol^{-1}
$\frac{2.303RT}{F}$ at 298 K	0.059 V	N_A	6.022×10^{23} mol^{-1}

Chapter 1 Solutions

Concentration terms

Mass % $= \dfrac{w_B}{w_{soln}} \times 100$; ppm $= \dfrac{w_B}{w_{soln}} \times 10^6$; $x_A = \dfrac{n_A}{n_A + n_B}$

Molarity $M = \dfrac{n_B}{V_{soln}\ (\text{L})}$ [mol L^{-1}]; Molality $m = \dfrac{n_B}{w_A\ (\text{kg})}$ [mol kg^{-1}] (temperature independent)

Laws

Henry: $p = K_H x$ (higher $K_H \Rightarrow$ lower solubility). Raoult: $p_A = x_A p_A^\circ$; $p_{total} = x_A p_A^\circ + x_B p_B^\circ$

Ideal: $\Delta_{mix}H = 0$, $\Delta_{mix}V = 0$. Positive deviation: $\Delta_{mix}H > 0$ (min-boiling azeotrope); negative: $\Delta_{mix}H < 0$ (max-boiling).

Colligative properties (with van't Hoff factor i)

RLVP $\dfrac{p^\circ - p}{p^\circ} = i x_B \approx i\dfrac{w_B M_A}{M_B w_A}$

Boiling point elevation $\Delta T_b = iK_b m,\ M_B = \dfrac{K_b \times w_B \times 1000}{\Delta T_b \times w_A}$

Freezing point depression $\Delta T_f = iK_f m,\ M_B = \dfrac{K_f \times w_B \times 1000}{\Delta T_f \times w_A}$

Osmotic pressure $\pi = iCRT = \dfrac{iw_B RT}{M_B V}$

$i = \dfrac{\text{normal molar mass}}{\text{abnormal molar mass}}$; dissociation $\alpha = \dfrac{i-1}{n-1}$; association $\alpha = \dfrac{1-i}{1-1/n}$

JEE extra

Vapour mole fraction $y_A = \dfrac{x_A p_A^\circ}{p_{total}}$. $\dfrac{1}{p_{total}} = \dfrac{y_A}{p_A^\circ} + \dfrac{y_B}{p_B^\circ}$. $K_b = \dfrac{RT_b^2 M_A}{1000\Delta_{vap}H}$. Isotonic: $i_1 C_1 = i_2 C_2$.

Chapter 2 Electrochemistry

Cells and Nernst equation

$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode} = E^\circ_{right} - E^\circ_{left}$ Notation: anode | anode soln ‖ cathode soln | cathode

$E_{cell} = E^\circ_{cell} - \dfrac{0.059}{n}\log Q$ (298 K); electrode: $E_{M^{n+}/M} = E^\circ - \dfrac{0.059}{n}\log\dfrac{1}{[M^{n+}]}$

$\Delta_r G^\circ = -nFE^\circ_{cell}$; $\log K_c = \dfrac{nE^\circ_{cell}}{0.059}$; $\Delta_r G^\circ = -2.303RT\log K_c$

Conductance

$R = \rho\dfrac{l}{A}$; $\kappa = \dfrac{1}{\rho} = \dfrac{G^*}{R}$, cell constant $G^* = \dfrac{l}{A}$ [cm^{-1}]; κ [S cm^{-1}]

$\Lambda_m = \dfrac{\kappa \times 1000}{C}$ [$\text{S cm}^2\ \text{mol}^{-1}$]; strong: $\Lambda_m = \Lambda_m^\circ - A\sqrt{C}$

Kohlrausch: $\Lambda_m^\circ = \nu_+\lambda_+^\circ + \nu_-\lambda_-^\circ$; weak: $\alpha = \dfrac{\Lambda_m}{\Lambda_m^\circ}$, $K_a = \dfrac{C\alpha^2}{1-\alpha}$

Electrolysis and batteries

Faraday: $m = ZIt = \dfrac{M}{nF}It$; $Q = It$; series cells: $\dfrac{m_1}{m_2} = \dfrac{E_1}{E_2}$ (equivalent weights)

Lead storage: $\text{Pb} + \text{PbO}_2 + 2\,\text{H}_2\text{SO}_4 \longrightarrow 2\,\text{PbSO}_4 + 2\,\text{H}_2\text{O}$ (discharge). Fuel cell: $2\,\text{H}_2 + \text{O}_2 \longrightarrow 2\,\text{H}_2\text{O}$.

Rusting (anode): $2\,\text{Fe} \longrightarrow 2\,\text{Fe}^{2+} + 4\,\text{e}^-$; cathode: $\text{O}_2 + 4\,\text{H}^+ + 4\,\text{e}^- \longrightarrow 2\,\text{H}_2\text{O}$.

Key E° (V): Li −3.05, K −2.93, Mg −2.37, Al −1.66, Zn −0.76, Fe^{2+} −0.44, H 0, Cu^{2+} 0.34, I_2 0.54, $\text{Fe}^{3+}/\text{Fe}^{2+}$ 0.77, Ag 0.80, Br_2 1.09, Cl_2 1.36, F_2 2.87.

JEE extra

Concentration cell $E = \frac{0.059}{n}\log\frac{C_2}{C_1}$. H electrode $E = -0.059$ pH. $E^\circ_3 = \frac{n_1E^\circ_1 + n_2E^\circ_2}{n_3}$. Λ°_m (weak) $= \Lambda^\circ$(salt) $+ \Lambda^\circ$(strong acid) $- \Lambda^\circ$(salt of strong acid).

Chapter 3 Chemical Kinetics

Rates

For $aA + bB \to cC$: rate $= -\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = \frac{1}{c}\frac{d[C]}{dt}$; rate $= k[A]^x[B]^y$, order $= x + y$

Units of k: $(\text{mol L}^{-1})^{1-n}\ \text{s}^{-1}$ (zero: mol L^{-1} s^{-1}; first: s^{-1}; second: L mol^{-1} s^{-1})

Integrated rate laws

	Rate law	Integrated	$t_{1/2}$
Zero	k	$k = \frac{[R]_0 - [R]}{t}$	$\frac{[R]_0}{2k}$
First	$k[R]$	$k = \frac{2.303}{t}\log\frac{[R]_0}{[R]}$	$\frac{0.693}{k}$

Gas phase (first order, $A \to B + C$): $k = \frac{2.303}{t}\log\frac{p_i}{2p_i - p_t}$. $t_{99\%} = 2t_{90\%}$; after n half-lives $[R] = [R]_0/2^n$.

Temperature

Arrhenius: $k = Ae^{-E_a/RT}$, $\ln k = \ln A - \frac{E_a}{RT}$ (slope $-\frac{E_a}{R}$); $\log\frac{k_2}{k_1} = \frac{E_a}{2.303R}\left(\frac{T_2 - T_1}{T_1T_2}\right)$

Fraction of effective collisions $= e^{-E_a/RT}$. A catalyst lowers E_a, does not change ΔG or K.

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Second order: $\frac{1}{[R]} - \frac{1}{[R]_0} = kt$, $t_{1/2} = \frac{1}{k[R]_0}$; nth order: $t_{1/2} \propto [R]_0^{1-n}$. Mean life $\tau = \frac{1}{k} = 1.44\,t_{1/2}$.

Chapter 4 The d- and f-Block Elements

Electronic configuration

d-block: $(n-1)d^{1-10}\,ns^{1-2}$ (Pd: $4d^{10}5s^0$). f-block: $(n-2)f^{1-14}(n-1)d^{0-1}ns^2$.

Exceptions (3d): Cr $= [Ar]3d^54s^1$, Cu $= [Ar]3d^{10}4s^1$ (half-filled / full d is extra stable).

Ions: remove 4s electrons *before* 3d. e.g. Fe $3d^64s^2 \to Fe^{2+}\ 3d^6$, $Fe^{3+}\ 3d^5$.

Transition element: has an incompletely filled d subshell in the atom *or* in a common ion. So Zn, Cd, Hg (d^{10} in atom and ions) are not transition elements; Cu (d^9 in Cu^{2+}) and Ag (d^9 in Ag^{2+}) are.

Magnetic moment and colour

Spin-only formula: $\mu = \sqrt{n(n+2)}$ BM (n = number of unpaired electrons)

n	1	2	3	4	5
μ (BM)	1.73	2.83	3.87	4.90	5.92

Paramagnetic — unpaired electrons present

Diamagnetic — all electrons paired (d^0, d^{10})

Colour: due to d–d transitions; seen colour = complementary to light absorbed. d^0 and d^{10} ions are **colourless**.

Sc^{3+}	$3d^0$	colourless	V^{2+}	$3d^3$	violet	Fe^{2+}	$3d^6$	green
Ti^{4+}	$3d^0$	colourless	Cr^{3+}	$3d^3$	violet	Co^{2+}	$3d^7$	pink
Ti^{3+}	$3d^1$	purple	Mn^{3+}	$3d^4$	violet	Ni^{2+}	$3d^8$	green
V^{4+}	$3d^1$	blue	Cr^{2+}	$3d^4$	blue	Cu^{2+}	$3d^9$	blue
V^{3+}	$3d^2$	green	Mn^{2+}	$3d^5$	pink	Zn^{2+}	$3d^{10}$	colourless
			Fe^{3+}	$3d^5$	yellow	Cu^{+}	$3d^{10}$	colourless

Trends across the 3d series (Sc to Zn)

Property	Trend	Exception and reason
Atomic radius	Decreases, then nearly constant	4d ≈ 5d size (Zr 160 pm, Hf 159 pm): lanthanoid contraction
Enthalpy of atomisation	High; maximum near middle (many unpaired e^-)	Zn lowest (126 kJ/mol): no unpaired d electrons
Ionisation enthalpy	Increases gradually, irregularly	Cr, Cu: high IE_2 (d^5, d^{10} broken); Mn, Zn: high IE_3
Oxidation states	Max number at Mn (+2 to +7)	Sc only +3; Zn only +2
$E^\circ(M^{2+}/M)$	Becomes less negative	Cu +0.34 V (high $\Delta_a H$, low $\Delta_{hyd} H$); Mn, Ni, Zn more negative than expected
$E^\circ(M^{3+}/M^{2+})$	Irregular	Mn +1.57 V, Co +1.97 V (strong oxidants); Sc, Zn: no M^{3+}/M^{2+} couple

Oxidation states and stability rules

- Highest oxidation state = group number (Sc to Mn): Sc_2O_3, TiO_2, V_2O_5, CrO_3, Mn_2O_7. Beyond Mn: Fe_2O_3 is highest oxide of Fe.
- Highest states are shown in **oxides and fluorides** (small, highly electronegative O and F; O forms multiple bonds). MnO_4^- (Mn^{+7}) vs MnF_4.
- Low oxide basic, high oxide acidic: MnO basic → Mn_2O_7 acidic; CrO basic, Cr_2O_3 amphoteric, CrO_3 acidic; V_2O_5 amphoteric.
- Stability of +2: increases across the first half (Mn^{2+} d^5 very stable).
- d^4 pair: Cr^{2+} reducing (→ d^3, half-filled t_{2g}); Mn^{3+} oxidising (→ d^5).
- Cu^+ unstable in water: $2\,Cu^+ \longrightarrow Cu^{2+} + Cu$ (high $\Delta_{hyd} H$ of Cu^{2+} outweighs IE_2).

Potassium dichromate, $K_2Cr_2O_7$

Preparation (from chromite ore $FeCr_2O_4$):

$$4\,FeCr_2O_4 + 8\,Na_2CO_3 + 7\,O_2 \longrightarrow 8\,Na_2CrO_4 + 2\,Fe_2O_3 + 8\,CO_2$$

$2\,Na_2CrO_4 + 2\,H^+ \longrightarrow Na_2Cr_2O_7 + 2\,Na^+ + H_2O$ $\quad$ $Na_2Cr_2O_7 + 2\,KCl \longrightarrow K_2Cr_2O_7 + 2\,NaCl$

pH effect: $2\,CrO_4^{2-}$ (yellow) $+ 2\,H^+ \rightleftharpoons Cr_2O_7^{2-}$ (orange) $+ H_2O$ (Cr is +6 in both)
Structure: CrO_4^{2-} tetrahedral; $Cr_2O_7^{2-}$ two tetrahedra sharing one O, Cr–O–Cr 126°.

Oxidising action (acid): $Cr_2O_7^{2-} + 14\,H^+ + 6\,e^- \longrightarrow 2\,Cr^{3+} + 7\,H_2O$ $\quad (E^\circ = 1.33\ V)$

with I^-: $Cr_2O_7^{2-} + 14\,H^+ + 6\,I^- \longrightarrow 2\,Cr^{3+} + 3\,I_2 + 7\,H_2O$

with Fe^{2+}: $Cr_2O_7^{2-} + 14\,H^+ + 6\,Fe^{2+} \longrightarrow 2\,Cr^{3+} + 6\,Fe^{3+} + 7\,H_2O$

with H_2S: $Cr_2O_7^{2-} + 8\,H^+ + 3\,H_2S \longrightarrow 2\,Cr^{3+} + 3\,S + 7\,H_2O$

with Sn^{2+}: $Cr_2O_7^{2-} + 14\,H^+ + 3\,Sn^{2+} \longrightarrow 2\,Cr^{3+} + 3\,Sn^{4+} + 7\,H_2O$

Potassium permanganate, $KMnO_4$

Preparation: $2\,MnO_2 + 4\,KOH + O_2 \longrightarrow 2\,K_2MnO_4$ (green) $+ 2\,H_2O$; $\quad 3\,MnO_4^{2-} + 4\,H^+ \longrightarrow 2\,MnO_4^- + MnO_2 + 2\,H_2O$
Commercial: alkaline oxidative fusion of MnO_2, then electrolytic oxidation of MnO_4^{2-}. Lab: $2\,Mn^{2+} + 5\,S_2O_8^{2-} + 8\,H_2O \longrightarrow 2\,MnO_4^- + 10\,SO_4^{2-} + 16\,H^+$
Heating: $2\,KMnO_4 \xrightarrow{513\,K} K_2MnO_4 + MnO_2 + O_2$. MnO_4^{2-} green, paramagnetic; MnO_4^- purple, diamagnetic; both tetrahedral.

Reduction half-reactions:

acidic: $MnO_4^- + 8\,H^+ + 5\,e^- \longrightarrow Mn^{2+} + 4\,H_2O$ $\quad (E^\circ = 1.52\ V)$

neutral/weakly alkaline: $MnO_4^- + 4\,H^+ + 3\,e^- \longrightarrow MnO_2 + 2\,H_2O$ $\quad (E^\circ = 1.69\ V)$

strongly alkaline: $MnO_4^- + e^- \longrightarrow MnO_4^{2-}$ $\quad (E^\circ = 0.56\ V)$

In acid:

I^-: $10\,I^- + 2\,MnO_4^- + 16\,H^+ \longrightarrow 2\,Mn^{2+} + 8\,H_2O + 5\,I_2$

Fe^{2+}: $5\,Fe^{2+} + MnO_4^- + 8\,H^+ \longrightarrow Mn^{2+} + 4\,H_2O + 5\,Fe^{3+}$

oxalate (333 K): $5\,C_2O_4^{2-} + 2\,MnO_4^- + 16\,H^+ \longrightarrow 2\,Mn^{2+} + 8\,H_2O + 10\,CO_2$

H_2S: $5\,S^{2-} + 2\,MnO_4^- + 16\,H^+ \longrightarrow 2\,Mn^{2+} + 8\,H_2O + 5\,S$

SO_3^{2-}: $5\,SO_3^{2-} + 2\,MnO_4^- + 6\,H^+ \longrightarrow 2\,Mn^{2+} + 3\,H_2O + 5\,SO_4^{2-}$

NO_2^-: $5\,NO_2^- + 2\,MnO_4^- + 6\,H^+ \longrightarrow 2\,Mn^{2+} + 5\,NO_3^- + 3\,H_2O$

In neutral / faintly alkaline:

$I^- \to IO_3^-$: $2\,MnO_4^- + H_2O + I^- \longrightarrow 2\,MnO_2 + 2\,OH^- + IO_3^-$

thiosulphate: $8\,MnO_4^- + 3\,S_2O_3^{2-} + H_2O \longrightarrow 8\,MnO_2 + 6\,SO_4^{2-} + 2\,OH^-$

Mn^{2+}: $2\,MnO_4^- + 3\,Mn^{2+} + 2\,H_2O \longrightarrow 5\,MnO_2 + 4\,H^+$

HCl is not used for $KMnO_4$ titrations (it is oxidised to Cl_2).

Other properties of transition metals

- **Catalysts** (variable oxidation states, complex formation): V_2O_5 (Contact process), Fe (Haber process), Ni (hydrogenation), $TiCl_4 + Al(CH_3)_3$ (Ziegler). Fe^{3+} catalyses $2\,I^- + S_2O_8^{2-} \longrightarrow I_2 + 2\,SO_4^{2-}$.
- **Interstitial compounds:** small atoms (H, C, N) in metal lattices, e.g. TiC, Mn_4N, Fe_3H, $TiH_{1.7}$; non-stoichiometric, high m.p., very hard, metallic conductivity, chemically inert.
- **Alloys:** atoms within about 15% in radius; steels (Cr, V, W, Mo, Mn), brass (Cu–Zn), bronze

(Cu–Sn).
- **Complexes:** small ions, high charge, vacant d orbitals, e.g. $[Fe(CN)_6]^{3-}$, $[Cu(NH_3)_4]^{2+}$.
- **Disproportionation:** one oxidation state goes to a higher and a lower one, e.g. $3\,MnO_4^{2-} + 4\,H^+ \longrightarrow 2\,MnO_4^- + MnO_2 + 2\,H_2O$.

Lanthanoids (La to Lu, 4f)

- Configuration: $[Xe]4f^{1-14}5d^{0-1}6s^2$; $Ln^{3+} = 4f^n$.
- **Main oxidation state +3.** Also +2: Eu^{2+} (f^7), Yb^{2+} (f^{14}), Sm^{2+} (reducing). +4: Ce^{4+} (f^0, $E^\circ = +1.74$ V, oxidant), Tb^{4+} (f^7, oxidant); Pr, Nd, Dy show +4 only in MO_2.
- **Lanthanoid contraction:** steady decrease in size from La to Lu, due to poor shielding by $4f$ electrons. Consequences: Zr ≈ Hf in size and properties (hard to separate); basicity of $Ln(OH)_3$ decreases La → Lu.
- **Reactions:** with dilute acids give H_2; burn in O_2 to Ln_2O_3; with C (2773 K) give Ln_3C, Ln_2C_3, LnC_2; with N_2 give LnN; $Ln(OH)_3$ basic.
- **Uses:** mischmetall (∼95% Ln, ∼5% Fe, traces S, C, Ca, Al) in Mg-alloys for bullets, shells, lighter flints; mixed oxides as cracking catalysts; phosphors.

Actinoids (Ac to Lr, 5f) and comparison

- Configuration: $[Rn]5f^{1-14}6d^{0-1}7s^2$. All radioactive.
- Oxidation states: common +3; max +4 (Th), +5 (Pa), +6 (U), +7 (Np, Pu); later members mainly +3.

	Lanthanoids	**Actinoids**
Orbitals	$4f$ (well shielded)	$5f$ (less shielded; $5f$, $6d$, $7s$ close in energy)
Oxidation states	Mainly +3	Wide range (+3 to +7)
Contraction	Smaller per element	Greater per element (poorer $5f$ shielding)
Radioactivity	Only Pm	All
Complexes	Fewer	More (higher charge/size ratio)

JEE extra

- **Equivalent weight** (redox titrations): $EW = \dfrac{M}{n}$; $KMnO_4$: $n = 5$ (acid), 3 (neutral), 1 (strongly alkaline); $K_2Cr_2O_7$: $n = 6$.
- **Normality equation:** $N_1V_1 = N_2V_2$. Mole ratios: $MnO_4^- : Fe^{2+} = 1:5$; $MnO_4^- : C_2O_4^{2-} = 2:5$; $Cr_2O_7^{2-} : Fe^{2+} = 1:6$.
- **Chromyl chloride test** (Cl^-): $K_2Cr_2O_7 + 4\,NaCl + 6\,H_2SO_4 \longrightarrow 2\,CrO_2Cl_2$ (red vapour) + ...
- **CrO_5** (blue, in ether): $Cr_2O_7^{2-} + 4\,H_2O_2 + 2\,H^+ \longrightarrow 2\,CrO_5 + 5\,H_2O$; Cr is +6 (butterfly peroxo structure).
- **Ionic radii trend:** Ln^{3+}: La^{3+} 103 pm → Lu^{3+} 86 pm.
- **Spin-only shortcut:** $\mu^2 = n(n+2)$; e.g. $\mu = 3.87 \Rightarrow n = 3$.

Chapter 5 Coordination Compounds

Terms and rules

Coordination number = number of donor atoms (en, ox = 2; edta = 6). Charge of entity = O.S. of metal + sum of ligand charges.

Ligands: unidentate (Cl^-, NH_3); didentate (en, $C_2O_4^{2-}$); ambidentate (NO_2^-/ONO^-, SCN^-/NCS^-, CN^-/NC^-).

Naming: cation first; ligands alphabetical then metal; anionic ligands -ido/-o (chlorido, cyanido, hydroxido, oxalato, nitrito-N); neutral: aqua, ammine, carbonyl, en; anionic complex: metal + -ate (ferrate, cuprate, argentate, aurate, plumbate, stannate); O.S. in Roman numerals.

Isomerism: structural (linkage, coordination, ionisation, solvate); stereo (geometrical cis/trans, fac/mer; optical).

VBT and CFT

CN 4	sp^3 tetrahedral / dsp^2 square planar	$[NiCl_4]^{2-}$ / $[Ni(CN)_4]^{2-}$
CN 6	d^2sp^3 inner orbital (low spin) / sp^3d^2 outer (high spin)	$[Fe(CN)_6]^{4-}$ / $[CoF_6]^{3-}$

Spectrochemical series: $I^- < Br^- < SCN^- < Cl^- < S^{2-} < F^- < OH^- < C_2O_4^{2-} < H_2O < NCS^- < edta^{4-} < NH_3 < en < CN^- < CO$

Octahedral: t_{2g} $(-0.4\Delta_o)$, e_g $(+0.6\Delta_o)$. $\Delta_o > P$: low spin; $\Delta_o < P$: high spin. $\Delta_t = \frac{4}{9}\Delta_o$.

$\mu = \sqrt{n(n+2)}$ BM; larger $\Delta_o \Rightarrow$ shorter λ absorbed. Colour = complementary of light absorbed.

JEE extra

EAN $= Z - \text{O.S.} + 2 \times \text{CN}$. Geometrical isomers: $[Ma_2b_2]$ planar 2; $[Ma_4b_2]$ 2; $[Ma_3b_3]$ 2 (fac, mer); [Mabcd] planar 3; $[M(AA)_2b_2]$ cis (optical pair) + trans; $[M(AA)_3]$ optical only.
Stability constant $\beta_n = K_1K_2\cdots K_n$; chelates have high β.